and in so doing, composed the score for what has come to be known as *classical physics.*

In the decades following Newton's work, his equations were developed into an elaborate mathematical structure that significantly extended both their reach and their practical utility. Classical physics gradually became a sophisticated and mature scientific discipline. But shining clearly through all these advances was the beacon of Newton's original insights. Even today, more than three hundred years later, you can see Newton's equations scrawled on introductory-physics chalkboards worldwide, printed on NASA flight plans computing spacecraft trajectories, and embedded within the complex calculations of forefront research. Newton brought a wealth of physical phenomena within a single theoretical framework.

But while formulating his laws of motion, Newton encountered a critical stumbling block, one that is of particular importance to our story (Chapter 2). Everyone knew that things could move, but what about the arena within which the motion took place? Well, that's space, we'd all answer. But, Newton would reply, what *is* space? Is space a real physical entity or is it an abstract idea born of the human struggle to comprehend the cosmos? Newton realized that this key question had to be answered, because without taking a stand on the meaning of space and time, his equations describing motion would prove meaningless. Understanding requires context; insight must be anchored.

And so, with a few brief sentences in his *Principia Mathematica*, Newton articulated a conception of space and time, declaring them absolute and immutable entities that provided the universe with a rigid, unchangeable arena. According to Newton, space and time supplied an invisible scaffolding that gave the universe shape and structure.

Not everyone agreed. Some argued persuasively that it made little sense to ascribe existence to something you can't feel, grasp, or affect. But the explanatory and predictive power of Newton's equations quieted the critics. For the next two hundred years, his absolute conception of space and time was dogma.